SatBhoomi Score + Raksha

Product Blueprint (Idea-to-Plan) - AI farm credit scoring and early-warning support for bank officers

Version / date	v0.1 draft, 26 October 2026
Mode	Idea-to-Plan with a student PoC target (Creative Intelligence for Engineers group project: Draft Paper + PoC)
How to read	Every item is tagged: **Verified** (source checked), **Customer** (you said it), **Assumption** (my starting choice, change if wrong), **Hypothesis** (needs testing).
Honest status	Nothing in this document has been built, run or tested. All test cases are planned, not executed.

1. Project Intake Summary

Item	Content
Raw idea (Customer)	SatBhoomi Score: a credit score for small farmers from satellite crop health, rainfall and mandi prices. Sir's feedback added: show what the platform does when a farmer cannot repay (module name: **Raksha**).
Known	Course deliverables are a Draft Paper + PoC. Team of 3. Sir asked to understand the bank loan process, documents, and how low-score farmers get loans. Bank rules checked (see section 3).
Unclear	District; how to get farm boundaries; team skills and roles; deadline; whether a bank or agri officer can be interviewed.
Assumed	Main user = bank officer, farmer sees a simple summary (Assumption). First crop = wheat, rabi season, because skies are clearer (Assumption). One district only (not chosen yet).
Needs validation	That crop health predicts repayment; that officers find what-if and relief evidence useful; that farm-level data can be obtained for a sample.
Top open question	Which district, and can you reach one real bank or agri officer for a 10-minute interview?
Next steps	Confirm district, run 3-5 short interviews, build the data pipeline first, then the model, then the dashboard (section 12).

2. Product Discovery

Problem statement

New-to-credit small farmers have no credit history, so banks struggle to judge them and may reject them or lend too much or too little. Farmers then use moneylenders. When a crop fails or prices crash, repayment stress builds up and, in the worst cases, becomes serious personal distress. Banks usually find out late, when an instalment is already missed.

Users and personas

Persona	Role in system	Main need	Pain today
Bank loan officer / branch manager (**primary**)	Runs assessment, decides the loan, follows up	Quick, explainable view of a farmer's farm risk; early warning in season	Few records for new farmers; field visits are slow; learns of trouble late
Farmer (secondary, read-only)	Receives a simple summary in his language	Understand why a limit was suggested and what helps	Does not know what the bank looks at
Branch admin / agri officer	Manages users, reviews alerts and reports	Portfolio view, evidence for relief requests	Manual paperwork

Value proposition

For bank officers lending to new farmers, SatBhoomi gives a transparent farm-based risk score with reasons, a right-sized loan suggestion, in-season stress alerts, and an evidence report to support early relief such as rescheduling. Unlike a plain approve or reject score, it tells the officer what to do next.

Scope boundaries

In scope (PoC)	Out of scope
One district, one crop (wheat), sample plots; score + reasons + what-if; replay of past seasons to test alerts; officer dashboard; farmer summary; relief evidence report; audit log	Real loan approval or rejection; integration with bank core systems; real farmer data without consent; multiple crops or states; a farmer mobile app; automatic recovery messages or calls; any prediction of a person's mental state

Initial feasibility (Hypothesis, medium confidence)

Technically doable for 3 students in about 4 weeks because satellite, rain and price data are public and a PoC can use sample plots. The hard parts are getting plot boundaries and having no real default labels. Both are handled in sections 7, 9 and 11.

Open question inventory

ID	Question	Why it matters	Status
Q1	Main user?	Decides screens	Answered by default: bank officer (Assumption)
Q2	First crop?	Decides data and season	Answered by default: wheat (Assumption)
Q3	Which district?	Data fetch, local checks	**Open**
Q4	How will plot boundaries be obtained?	Needed for farm-level NDVI	**Open** (fallback: draw sample polygons or use village-level pixels)
Q5	Can we interview a bank or agri officer?	Validation and paper quotes	**Open**
Q6	Team skills and roles?	Task split	**Open**
Q7	Submission deadline?	Roadmap length	**Open**

3. Market and Business Validation

Important finding (Verified): satellite-based farm credit scoring is not a new idea. SatSure, a Bengaluru company, already sells it to banks. So do not claim 'first' or 'unique' in the paper. Cite SatSure as related work and win on the differences below.

What exists (Verified, sources at end of this section)

- SatSure's SatSource platform describes a farm score and farm report with verified boundaries, six seasons of cropping history, crop identification, irrigation condition, rainfall and groundwater stress, and an interpretable risk view. It is delivered to banks through APIs. Covers 630,000+ plots in 230+ districts.
- SatSure Sage monitors loans through their life cycle and supports collection planning. SatSure also launched a joint product with TransUnion CIBIL. ICICI Bank, Kotak Mahindra Bank and HDFC Ltd. are listed as investors.
- Government and bank context: the collateral-free limit for agri loans is Rs. 2 lakh (RBI), and relief such as rescheduling is allowed when crop loss is 33% or more (RBI relief guidelines).

Comparison and differentiation

Capability	SatSure (public pages only)	SatBhoomi PoC (planned)
Farm-level risk score for lending	Yes (Verified)	Yes, proxy-label model on sample plots
In-season monitoring after loan	Yes, via Sage (Verified)	Yes, replayed on past seasons
Reasons behind a score	'Interpretable risk view' (Verified wording)	Per-farmer top reasons (SHAP) in plain words
What-if simulator (e.g. add insurance or irrigation)	Not found in pages we read; **unverified**	Planned (Hypothesis: officers will find it useful)
Relief evidence report matched to RBI 33% rule	Not found; **unverified**	Planned (supporting evidence only, official assessment stays with authorities)
Farmer-facing summary in Gujarati or Hindi	Not found; **unverified**	Planned
Scale and data quality	Large, commercial data	One district, sample plots, free data

Positioning for the paper

An open, explainable, student-built decision-support prototype that goes from 'score' to 'action': right-sized loan, what-if, early warning and relief evidence. Say clearly that SatSure's full feature set is not public, so the differences are hypotheses to test, not claims.

Key assumptions and validation plan

Assumption	Test	Success criterion (proposed, needs approval)
Officers want reasons and what-if, not only a score	Show clickable demo to 3-5 officers or agri officers	At least 3 of 5 rate usefulness 4 or more out of 5
Crop health carries real repayment signal	Train on past seasons, test on the newest season	AUC at least 0.70 on the held-out season and beats the simple baseline
Early alerts arrive in time to act	Replay past seasons; measure weeks between first alert and yield loss	Alert comes before harvest in most stressed cases
Relief report is credible	Compare its loss band with known bad seasons in the district	Matches known bad seasons

Recommendation: **GO, with revised positioning** (not 'novel score', but 'score to action'). Revisit after interviews.

Sources (accessed 26 Oct 2026): aikosh.indiaai.gov.in SatSource use case; satsure.co (investors, Sage, Sparta, CIBIL product); frontiertech.niti.gov.in SatSure story; rbi.org.in relief measures circular (33% rule); RBI KCC Directions 2026 summary (taxguru.in). Marketing pages show claims, not independent audits.

4. Requirements

Functional requirements

ID	Requirement	Actor	Pri.
FR-001	Officer logs in; roles: officer, admin	Officer	P0
FR-002	Create or select a farm plot: district, village, crop, area, location or polygon, irrigation type, insured yes/no	Officer	P0
FR-003	Load season data for the plot: crop health (NDVI), rainfall, mandi price, district yield	System	P0
FR-004	Compute risk score 0-100 and band (low / medium / high)	System	P0
FR-005	Show top reasons for the score in plain words	Officer	P0
FR-006	Suggest loan structure: limit band (from scale of finance), repayment month, insurance flag	Officer	P0
FR-007	What-if: change one controllable input and re-run the score	Officer	P0
FR-008	In-season monitoring: raise stress alerts (crop health drop, rain deficit, price crash)	System	P0
FR-009	Alert list with action playbook (visit, insurance claim help, extension) and a log of action taken	Officer	P1
FR-010	Relief evidence report: crop health before vs after, loss band vs 33% threshold, with disclaimer	Officer	P1
FR-011	Farmer summary in English, Gujarati, Hindi (templates first, LLM wording later with human review)	Officer	P1
FR-012	Audit log of every score, what-if and officer action	System	P1
FR-013	Record farmer consent before using real data	Officer	P1
FR-014	Export assessment as PDF	Officer	P2

Non-functional requirements (all targets are recommendations until approved)

Area	Target
Performance	Score for one plot in under 10 s when data is cached; pages load in under 3 s
Reliability	If satellite data is missing (clouds), show 'data gap' and use the last good reading; never crash or invent values
Security and privacy	Role-based access, hashed passwords, HTTPS, no secrets in code, audit log, synthetic data in the public demo
Explainability	Every score shows its top 3 reasons; every what-if shows old and new score
Usability	Works on phone and laptop; large text; Gujarati font supported; simple language
Reproducibility	Fixed random seeds, versioned data snapshot, model version stored with each score

User stories (examples)

Story	Acceptance criteria (Given / When / Then)
As a loan officer, I want a score with reasons, so that I can explain my decision.	Given a saved plot with data, when I open Assessment, then I see a 0-100 score, a band and 3 reasons within 10 s.
As a loan officer, I want to try insurance or drip irrigation, so that I can advise the farmer.	Given a result, when I change insurance to Yes, then the new score and the change are shown and the original is kept.
As a branch manager, I want an early alert when crop health drops, so that I can act before default.	Given a monitored plot, when crop health falls below the plot's own baseline by the set threshold, then an alert appears with suggested actions.
As a branch manager, I want a relief evidence report, so that I can support a rescheduling request.	Given an alert, when I open the report, then I see before/after crop health, an estimated loss band and a note that official assessment decides.

5. UX and Interface Specification

Screen inventory

ID	Screen	Purpose	Key states
S1	Login	Authenticate officer	Error: wrong password; locked after repeated failures
S2	Dashboard	Count of plots, open alerts, recent assessments	Empty: 'No plots yet, add one'; loading skeleton
S3	Add / select plot	Enter plot details; pick on map or enter coordinates	Validation: area, district, crop required; success toast
S4	Assessment result	Score, band, top reasons, loan suggestion, data-gap warnings	Loading while data fetched; data-gap banner; error with retry
S5	What-if	Change one input, see new vs old score	Validation on allowed inputs only
S6	Monitoring and alerts	Season timeline of crop health and rain; alert list	Empty: 'No alerts'; severity colours plus text labels
S7	Alert detail and playbook	Suggested actions, log what was done	Success: action saved
S8	Relief evidence report	Before/after view, loss band, disclaimer, print/PDF	Disclaimer always visible
S9	Farmer summary	Short plain-language summary, language switch	Fallback to template text if LLM unavailable

Main flows

- **Assess a new farmer:** S1 Login, S2 Dashboard, S3 Add plot, S4 Result, S5 What-if, S9 Farmer summary.
- **Handle stress during the season:** S2 Dashboard, S6 Alerts, S7 Playbook, S8 Relief report.

Design direction

- Calm green and navy palette, high contrast, large numbers for the score, colour never the only signal (always a text label).
- Mobile first layout; breakpoints around 600 px and 1000 px; Noto Sans Gujarati and Devanagari fonts for local language text.
- Keyboard friendly forms with labels; clear error text next to the field; plain words, no jargon.
- Wording rule for Raksha: call it 'repayment stress' and 'support needed', never anything that labels a person's mental state.

Status: this is a text specification only. Wireframes have not been drawn yet (next step: simple Figma or paper sketches of S4 and S6, then validate with 3-5 officers).

6. System Architecture and Technology

Style: modular monolith (one codebase, clear modules). Microservices are not justified for a 3-person PoC.

Data connectors Satellite NDVI, rainfall, mandi prices, district yield	**Feature pipeline** Cleans data, builds season features, caches	**Risk module** Baseline rules + ML model + SHAP reasons
Alert engine Rule-based stress alerts, playbook	**API layer** (FastAPI) Auth, plots, assessments, alerts, reports	**Dashboard** (Streamlit) Officer screens S1-S9
Database SQLite (PoC), PostgreSQL later	**Summary generator** Templates; optional LLM wording, human reviewed	**Report generator** Relief evidence and PDF export

Trust boundaries: public data providers (untrusted input, validate everything), the app and database (trusted), the LLM provider (external, receives only non-identifying text).

Architecture decisions (ADRs)

ID	Decision	Reason
ADR-1	Modular monolith	Small team, easy to run and demo
ADR-2	Rules baseline first, ML second	Shows the model adds value; protects against a failed model
ADR-3	Streamlit for PoC UI, React later if needed	Fastest path to a working dashboard
ADR-4	SQLite for PoC, schema written so PostgreSQL works	Zero setup now, easy upgrade
ADR-5	LLM only rewrites approved text into local language; never decides	Keeps decisions explainable and human

Technology choices (check current terms and pricing at setup)

Part	Recommended	Alternative	Trade-off
Language	Python 3.11	-	Best ML ecosystem; team likely knows it
Satellite data	Sentinel-2 NDVI via Google Earth Engine (free for non-commercial use; confirm registration)	Copernicus Data Space or Microsoft Planetary Computer	Earth Engine is easiest; account approval can take time, so start early
Rainfall	CHIRPS or NASA POWER	IMD gridded data	Free and global; IMD is local but formats vary
Mandi prices	Agmarknet data via data.gov.in	Manual CSV download	API key needed; data gaps possible
ML	scikit-learn, XGBoost, SHAP	LightGBM	Well known, explainable
Backend / UI	FastAPI + Streamlit	Flask + React	Fast build vs more polished UI
Database	SQLite then PostgreSQL	-	No server now; migrate later
Quality / CI	pytest, GitHub Actions	-	Free for student repos
Hosting	Local demo, or a free-tier host	Small cloud VM	Free tiers have limits; verify before the demo

7. Data Model

Entity	Key fields	Relationships
user	id, name, role, password_hash, created_at	1 user - many audit_log rows
farmer	id, display_name (or code), village, district, consent_id	1 farmer - many plots
plot	id, farmer_id, crop, area_acre, geometry or lat/lon, irrigation, insured	1 plot - many seasons, assessments
season_observation	id, plot_id, season, ndvi_mean, ndvi_peak, ndvi_std, rain_total_mm, rain_deficit_pct, price_mean, yield_district	many per plot (unique plot + season)
assessment	id, plot_id, model_version, score, band, reasons_json, loan_suggestion_json, created_by, created_at	1 assessment - many what_if rows
what_if	id, assessment_id, changed_input, old_value, new_value, new_score	-
alert	id, plot_id, type, severity, detected_on, status	1 alert - many action_log rows
action_log	id, alert_id, action_type, note, done_by, done_on	-
consent	id, farmer_id, purpose, given_on, withdrawn_on	-
audit_log	id, user_id, event, entity, entity_id, at	-

Example schema (excerpt)

```
CREATE TABLE plot (id INTEGER PRIMARY KEY, farmer_id INTEGER NOT NULL REFERENCES farmer(id), crop TEXT NOT NULL, area_acre REAL NOT NULL CHECK(area_acre > 0), irrigation TEXT, insured INTEGER DEFAULT 0, lat REAL, lon REAL);
CREATE TABLE season_observation (id INTEGER PRIMARY KEY, plot_id INTEGER NOT NULL REFERENCES plot(id), season TEXT NOT NULL, ndvi_mean REAL, ndvi_peak REAL, rain_total_mm REAL, price_mean REAL, UNIQUE(plot_id, season));
CREATE INDEX idx_obs_plot ON season_observation(plot_id);
```

Data lifecycle: store derived season features, not raw satellite images; delete a farmer's data on request; use synthetic or anonymised farmers in any public demo; never store ID numbers or bank account numbers.

8. API Contract (v1, JSON)

Method and path	Purpose	Auth	Notes
POST /api/v1/auth/login	Get session token	None	Rate limited; generic error text
POST /api/v1/plots	Create plot	Officer	Validates area, crop, district
GET /api/v1/plots?limit=&offset=	List plots	Officer	Pagination, filter by district
POST /api/v1/plots/{id}/assessments	Compute score, reasons, loan suggestion	Officer	Stores model version; idempotency key
GET /api/v1/assessments/{id}	Read an assessment	Officer	-
POST /api/v1/assessments/{id}/what-if	Run a what-if	Officer	Only allowed inputs
GET /api/v1/plots/{id}/monitoring	Season timeline and alerts	Officer	Includes data-gap flags
POST /api/v1/alerts/{id}/actions	Log an action taken	Officer	Audited
GET /api/v1/plots/{id}/relief-report	Evidence report	Officer	Disclaimer included
GET /api/v1/assessments/{id}/farmer-summary?lang=gu	Local language summary	Officer	Fallback to template

Error format: {"error": {"code": "VALIDATION_ERROR", "message": "area_acre must be above 0", "field": "area_acre"}}. Status codes: 200, 201, 400, 401, 403, 404, 429, 500. For the PoC, Streamlit may call the service layer directly, but keep these contracts so a React app or bank system can plug in later.

9. AI / Machine Learning Specification

Is ML really needed?

Start with a transparent rules baseline (for example: low average crop health plus rain deficit equals high risk). ML is justified only if it beats this baseline on held-out data. If it does not, the paper should say so honestly.

Label (the hardest part)

No real loan default data is public. The PoC uses a **proxy label**: a plot-season is 'stressed' when the district yield or the plot's peak crop health falls well below its own multi-season trend (threshold to be set and documented, Assumption). Risk: if the label is built from the same inputs as the features, the model will look falsely accurate (label leakage). Mitigation: build the label from yield or outcome data, and features only from earlier in the season.

Item	Specification
Features	Mean and peak NDVI, NDVI variation, early-season NDVI, rainfall total and deficit, number of past stressed seasons, price volatility, area, irrigation, insured flag
Models	Rules baseline, logistic regression, random forest, XGBoost; pick the simplest that performs well
Evaluation	Time-based split (train on older seasons, test on the newest); AUC, PR-AUC, calibration; check results by plot size; report confidence intervals if possible
Explanations	SHAP values turned into 3 plain-language reasons per score
What-if	Re-runs the model with one changed input. Important: this is a model-based estimate, not proof that the action causes the change. Label it 'estimate' on screen and in the paper
Score to band	0-100 scale with bands set after calibration; store model version with each score
LLM use	Only to rewrite an already approved template explanation into Gujarati or Hindi. No farmer identifiers sent. Human review of translations before real use. Fallback: template text
Human control	The system never approves or rejects a loan. The officer decides and the decision is logged
Monitoring	Track data-gap rate, alert counts, score drift between seasons

Raksha (early warning) design principles

- Alerts describe **farm and repayment stress** (crop health, rain, price), not a person's mental state. No individual-level prediction of self-harm.
- Every alert leads to a **human, supportive** action first: visit, help with insurance claim, request for rescheduling where rules allow, repayment timing linked to sale. Recovery pressure is not an action in the playbook.
- Relief evidence supports a request. The official crop loss assessment and calamity declaration stay with government and bank committees.
- If an officer meets a farmer in visible distress, the playbook tells them to involve local support and a senior colleague and to avoid handling it alone.

10. Security, Privacy and Compliance

Asset / area	Threat	Mitigation
Farmer identity and land details	Unauthorised access or leak	Role-based access, minimal fields, no ID or bank numbers, HTTPS, encrypted storage where available
Accounts	Weak passwords, brute force	Hashed passwords (bcrypt or argon2), rate limit, lockout
Scores and alerts	Misuse for harsh recovery or unfair denial	Human decision always, wording rules, audit log, supportive playbook only
Training data and model	Bad or poisoned data; silent drift	Validate inputs, version datasets and models, monitor drift
External data and LLM	Untrusted input; data leakage	Treat provider data as untrusted; send only non-identifying text to the LLM
Secrets	API keys committed to repository	Environment variables, secret scanning in CI
Demo	Real farmers exposed	Synthetic or anonymised sample farmers only

Legal note: India's Digital Personal Data Protection Act, 2023 is likely relevant once real personal data is used (consent, purpose limits, deletion). This is not legal advice and has not been verified for your case. For the course PoC, use synthetic data or data with written consent, and ask your instructor or a bank contact before touching real farmer records.

11. Risk Register

ID	Risk	Impact	Mitigation
R1	Idea seen as not new because SatSure exists	High	Cite them; position as score-to-action; validate differences with interviews
R2	No real default labels	High	Proxy label, state it openly, avoid label leakage
R3	Farm boundaries unavailable	High	Use sample polygons or village-level pixels; say so in limitations
R4	Cloud gaps in images	Medium	Wheat season, data-gap handling, interpolation flagged on screen
R5	Earth Engine or API access delays	Medium	Apply on day 1; fallback to Copernicus or Planetary Computer
R6	Time overrun (4-week plan)	Medium	P0 first; cut P1/P2 if late
R7	Ethical misuse of alerts	High	Wording rules, human-in-loop, supportive playbook
R8	Over-claiming accuracy in the paper	Medium	Report held-out results only; list limitations

12. Implementation Backlog and Roadmap

Estimates are rough (low confidence) and assume 3 students working part-time. No guaranteed dates.

Task	Description	Links	Depends	Done when
T-01	Repo, environment, folder structure, CI with pytest	-	-	CI runs a sample test
T-02	Satellite NDVI fetch for sample plots, 5 seasons	FR-003	T-01	CSV of NDVI per plot-season
T-03	Rainfall and price loaders; yield data	FR-003	T-01	Joined dataset saved
T-04	Features and proxy label; leakage check	FR-004	T-02, T-03	Documented label rule
T-05	Rules baseline + ML models + time-split evaluation	FR-004	T-04	Metrics table
T-06	SHAP reasons and plain-language text	FR-005	T-05	3 reasons per score
T-07	Loan suggestion rules (limit band, month, insurance)	FR-006	T-05	Rule tests pass
T-08	What-if engine	FR-007	T-05	Old vs new score shown
T-09	Alert engine and season replay	FR-008	T-04	Alerts on past bad seasons
T-10	Database and API layer	FR-001, FR-002	T-01	Endpoints tested
T-11	Dashboard screens S1-S6	FR-001 to FR-009	T-06 to T-10	Demo flow works
T-12	Playbook, relief report, farmer summary	FR-009 to FR-011	T-09, T-11	S7-S9 working
T-13	Audit log, consent, security checks	FR-012, FR-013	T-10	Checklist complete
T-14	Interviews, paper draft, demo rehearsal	-	T-11	Paper and demo ready

Milestones: Week 1 data and label (T-01 to T-04); Week 2 model and explanations (T-05 to T-08); Week 3 alerts, API, dashboard (T-09 to T-12); Week 4 security, interviews, paper, demo (T-13, T-14). Interviews should start in Week 1, not Week 4.

13. Test Strategy (planned, none executed yet)

Test	Requirement	Scenario	Expected	Status
TC-001	FR-001	Wrong password 5 times	Locked, generic error	Not run
TC-002	FR-002	Area = 0 or empty crop	Validation error on field	Not run
TC-003	FR-003	Satellite data missing for a season	Data-gap banner, no crash	Not run
TC-004	FR-004	Same plot scored twice	Same score (deterministic)	Not run
TC-005	FR-005	Score shown	3 reasons present	Not run
TC-006	FR-007	Change insurance to Yes	New and old scores shown	Not run
TC-007	FR-008	Replay a known bad season	Alert raised before harvest	Not run
TC-008	FR-010	Open relief report	Disclaimer visible; loss band shown	Not run
TC-009	FR-012	Any score or action	Audit row created	Not run
TC-010	Security	Officer opens admin route	403	Not run
TC-011	ML	Time-split evaluation	AUC at least 0.70 and beats baseline (target)	Not run

Traceability (summary): FR-004 to FR-007 map to screens S4 and S5, endpoints for assessments and what-if, and tests TC-004 to TC-006 and TC-011. FR-008 to FR-010 map to S6 to S8, monitoring and relief endpoints, tests TC-007 and TC-008.

14. Deployment, Operations and Cost

Area	Plan
Environments	Local development; one demo environment (local laptop or a free-tier host)
CI/CD	GitHub Actions: lint, tests on each push; tag a release before the demo
Secrets	Environment variables, never in the repository
Backups and rollback	Export the database file and model files before the demo; rollback = previous git tag
Logging and health	Application log with no personal data; simple health check route
Runbook (demo day)	Start app, load sample data, run the 2 main flows, keep a recorded video as backup

Cost estimate (Estimate, provider prices not verified)

Scenario	Assumption	Expected cost
Conservative	Local run, free data sources, templates only	About zero
Expected	Free-tier hosting, a few hundred short LLM translations	Low; check the provider's current price page
High	Paid host plus larger LLM use for a longer pilot	Not estimated; revisit if a pilot is approved

Main effort is student time: roughly 4 weeks part-time for 3 people (Assumption, low confidence). Scaling beyond one district would need paid satellite or boundary data, a proper database, and a bank partnership.

15. Quality Gates and Next Actions

Gate	Status	What is missing
A Discovery	Drafted	District, team roles, deadline
B Validation	Partly done	Competitor check done; interviews and criteria approval pending
C Requirements	Drafted	Sir or team approval of priorities
D Design	Drafted (text only)	Wireframes for S4 and S6
E Implementation readiness	Partly	Data access confirmed; label rule agreed
F MVP verification	Not started	Everything is planned, nothing executed
G Release readiness	Not started	-

Recommended next actions

- Pick the district and apply for satellite data access today (access can take time).
- Book 3-5 short interviews (bank officer, cooperative bank, agri extension officer) using the 5 questions from our earlier discussion; even phone calls count.
- Agree the proxy label rule and write it in one paragraph for the paper.
- Build T-02 to T-04 first. A working dataset makes everything else easier.
- Add SatSure to the related-work section and state your differences as hypotheses.